Semantic reads in `hymeko_query`: bases, is-a and field provenance

Created 2026-09-30 21:05 CEST Estimate: about 1–1.5 h

Decision. On 2026-09-30 the user decided DS0 of core `docs/plans/2026-09-30-core-after-extraction`: "Ds0 → query". Borrowed semantic queries over the canonical IR belong to `hymeko_query`, not Core. Core keeps the IR and its identity. This moves the reduced S1/S2 of that plan here, and no Core edit or approval slug is needed.

Evidence.

- Effective fields already live here: `fields.rs`, `effective_child` and `effective_value`. Geometry and webxr read them.
- The is-a walk is written again at least eleven times across repositories. In this repository alone:
  - `engine.rs check_inheritance`: depth 8, not reflexive;
  - `predicate_expr.rs` and `rewrite/template.rs`: by name, depth 16;
  - webxr `isa_transitive`: reflexive, depth 1024, a cycle reads as "not a".
- webxr's own-slot rule (`query.rs own`) is crate-private. HIVE and robotics check only that a slot is a leaf.

API (additive). In a new `hymeko_query/src/bases.rs`:

- `BasePolicy::{All, Tagged(SymId)}` says which base relations count. A tagged base carries the tag symbol on its reference annotation, as `<isa>` does. The tag is a symbol, not a string, so the walk needs only the IR: webxr's validators hold no name table.
- `bases(ir, decl, policy)`: the direct bases, in authored order. `QueryEngine` forwards to the free functions.
- `is_a(ir, decl, kind, policy) -> Result<bool, BaseError>`:
  - reflexive, and without a depth limit;
  - a diamond gives one answer; the full walk visits a shared base once, and the shallow walk below may revisit it within its visit bound;
  - a cycle on the walk is `BaseError::Cycle`, and a base outside the declaration table is `BaseError::Dangling`.

In `fields.rs`:

- `FieldBinding { declaration, provider }` and `effective_binding(decl, key)`. `effective_child` becomes a thin reader over it, so the lookup rule has one implementation.
- `own_child(decl, key)`: the declaration's own child only, the rule a writer uses for a slot.

This slice migrates.

- webxr `isa_transitive` and `each_base` use `is_a(..., All)`. A cycle still reads as "not a", which is today's result once the budget runs out.
- webxr `own` uses `own_child`.
- A diamond fixture (S4) records the current depth-first, authored-order rule for `effective_binding`.

Not in this slice.

- Query's own walks: `InheritsFromId` (depth 8), `predicate_expr` and `template`. They change behaviour and wait for DS1/DS2.
- Geometry, dynamics, optimize, network, `hymeko_wasm` and the C port.
- Graph (`hymeko_pgraph IsaWalk`) keeps its own `<isa>` walk. Transformation depends on graph, so graph cannot depend on this crate without a repository cycle.

Open decisions.

- DS1: the policy each consumer uses. The code is split: graph and network count only `<isa>`, query and webxr count every base.
- DS2: keep depth-first authored order, or C3, for effective fields. The S4 fixture pins the current rule until then.

Implementation record (2026-09-30).

- `is_a` first tries an allocation-free walk, with its path on the call stack, up to depth 64 and 512 visits. Beyond that it falls back to the walk that remembers finished declarations. A test checks that both give the same answer for every pair of declarations and both policies, cyclic IR included; another covers a 100-deep chain and a 12-diamond ladder.
- webxr validation bench (release, 100k entities): 34–36 ms before and 50 ms after. Parse and lowering take about 2 s. The first cut allocated on every call and took 212 ms; the shallow walk removed most of that. The rest pays for the new dangling-base check.
- Tests:
  - `hymeko_query`: 34 unit and 220 integration;
  - `hymeko_webxr`: 13 unit and 28 integration;
  - transformation workspace: 612 passed. The only failure is the known emitter fingerprint.
- Kawamirai with this webxr (local patch): 55 passed, as with the pinned revision.
- Duplication is 385/6059, as at the base. The query tests share `test_support`.

Verification.

- New tests for bases, is-a (reflexive, diamond, cycle, dangling, tagged), binding provenance, `own_child` and the diamond fixture.
- The `hymeko_query` and `hymeko_webxr` suites.
- The transformation workspace. The emitter fingerprint failure is known and already on the base.
- Kawamirai, which binds visuals through webxr, and `cargo tree -d`.
- Duplication must not rise.

